Cloud-Based Stock Market Analytics Dashboard

By Johnathan Nguyen | CTS2375 – Cloud Infrastructure and Services | Project Proposal

1. Project Summary

This project proposes an interactive web dashboard that can be used to explore historical data about stocks and exchange-traded funds (ETFs). The project will be built on a previous stock market project that will combine a PostgreSQL-backed Python analytics system with searchable price charts and a keyword-based stock query assistant. Once the application has been completed, it will run on an Amazon EC2 instance, as well as PostgreSQL and Amazon S3, which will hold processed datasets and database backups. The goal will be to have a working dashboard allowing people to analyze historical data.

2. Problem and Intended Users

The main problem that I am trying to tackle is that historical stock records can be difficult for students and beginning investors to compare directly, as individual price records and raw tables do not clearly show changes over time, differences in return or volatility, or the answer to questions such as “Which stock was most volatile during this period?” This will make the intended users students and anyone who may be interested in exploring historical market behavior. The dashboard will make stock selection, data filtering, quantitative comparisons, and explanations available within one interface.

3. Objectives and Measurable Success Criteria

In order for the product to be considered complete, it will need to at least run a browser-accessible dashboard on EC2, read historical stock records from PostgreSQL that will be hosted on EC2, use S3 for processed data uploads and recoverable backups, and display a selected stock's historical chart and statistics. In terms of testing these, I will verify by testing whether stock selection, date filtering, and the data queries work. I will also have an end goal of being able to display a typical analysis within five seconds and to test whether the stock query assistant works by using set equations and comparing it to other calculations

4. Workflow and System Architecture

For the workflow and system architecture, I need to ensure that it deploys properly. To do so, the first thing that will be done is to clean and transform the historical stock and ETF files using Python, which can then be uploaded into PostgreSQL, where I will process the data that will be privately stored in S3. As for the UI, the user will need to open the Streamlit website and use a ticker on the left side to search between stocks and ETFs, which will then let them select a date range and view metrics in the central panel. The right side will be used for the assistant, which can help interpret questions to help the user understand things based on the predefined keywords and calculations. The browser will send requests to Streamlit on EC2, while the application code queries PostgreSQL on the same instance and will use Pandas to calculate results. It will then return Plotly charts and text.

5. Technology and Cloud-Service Justification

For this project, it will use Infrastructure as a Service (IaaS), which will be provided by Amazon EC2. This will allow control over the Python web application and PostgreSQL process within one virtual machine while reducing the complexity for a small project. S3 will then provide managed object storage for portable processed CSVs and database backups. I chose Streamlit because it will allow the dashboard

to be built using Python, while Plotly will give us interactive charts. PostgreSQL will then be used to organize the stock records, while pandas will be used to make the financial calculations easier to perform.

6. Security, Reliability, Monitoring, and Cost

As for security, the EC2 instance will use an IAM role to access only the S3 resources that it will need, making it so that AWS access keys won't be written into the code. I will also make sure the S3 bucket will stay private and that PostgreSQL only accepts connections from the application on the instance. Another security level would be having a security group that will have a rule to control who has access and limit people's administrative access. And since everything will run on one EC2 instance, the dashboard will become unavailable if the instance stops. For monitoring, I will use basic CloudWatch metrics and application logs to check for problems. As for costs, I will use a small instance and stop it when I don't need it, as well as setting budget alerts to ensure costs stay low. If the project is lost, I will use dataset copies and database backups that will be stored in S3 to restore it.

7. Implementation Plan and Evaluation

For the project, I will be working alone, meaning I'll be working on the code, cloud setup, testing, as well as the documentation. As for the plan for this project, I plan to clean the dataset, check the PostgreSQL database, then build the financial calculations, as well as test the keyword-based assistant, create the UI dashboard, and connect the application to EC2 and S3. For the evaluation, I will show the dashboard, successful database queries, and that file transfers work, and I will test the results and include screenshots.

8. Risks, Fallback, and Demonstration

Some problems that I may run into will be missing or inconsistent stock data, questions that the assistant might not understand, AWS setup errors, as well as unexpected costs. To reduce these risks, I plan on checking the dataset and cleaning it, making the supported questions clear, manually testing the calculations, and using budget alerts to ensure low cost. If a problem occurs on the cloud side of things, I can always use local versions to fix the issue and carry on with development. As for the demonstration, I will select a stock and show its historical price chart, ask the algorithm about volatility or return, and show that the application is running on EC2 and can be downloaded or uploaded using a file from S3.

Works Cited

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